

3.1 IDE & CLI

An **Integrated Development Environment (IDE)** is a full graphical application providing a complete set of tools for software development in one package, while a **Command Line Interface (CLI)** is a text-based application where users type commands to execute tasks.

CLI (Command Line Interface)

CLIs are used for speed, automation, scripting, and low resource consumption. Users interact purely with text commands.

Examples of CLI environments/shells:

- **Bash** (common on Linux and macOS)
- **Command Prompt (cmd.exe)** (traditional Windows CLI)
- **PowerShell** (modern, more powerful Windows CLI)
- **Terminal** (standard on macOS)

Examples of common CLI tasks/commands:

- **File Management:**
 - `ls` or `dir` to list files in a directory.
 - `cd [directory_name]` to change the current directory.
 - `mkdir [new_directory]` to create a new directory.
- **Version Control:**
 - `git clone [repository_url]` to copy a remote repository to your local machine.
 - `git commit -m "[message]"` to save changes to the local repository.
- **Building/Packaging:**
 - `npm install` to install project dependencies (Node.js projects).
 - `maven clean install` to build a Java project using Maven.

IDE (Integrated Development Environment)

IDEs integrate multiple features into a single graphical user interface, making code navigation, debugging, and project management easier.

Examples of popular IDEs:

- **Visual Studio** (primarily for C#, C++, .NET development)
- **IntelliJ IDEA** (popular for Java development, part of the JetBrains suite)
- **Eclipse** (open-source, widely used for Java and other languages with plugins)
- **PyCharm** (specifically designed for Python development)
- **Xcode** (for macOS and iOS application development)

Examples of common IDE features/workflows:

- **Code Editing:** Provides smart features like syntax highlighting, error squiggly lines, and intelligent code completion (autocomplete).
- **Debugging:** Allows developers to set breakpoints visually and step through code execution line by line while inspecting variable values.
- **Integrated Tools:** Includes built-in version control integration (e.g., Git GUI tools), testing frameworks, and dependency management.
- **Refactoring:** Provides automated tools for safely renaming variables, functions, or classes across an entire project.

Many modern IDEs also include built-in terminal emulators, allowing users to leverage the power of the CLI within the same graphical environment.